EE5203 L06 Lab Guide

Button interrupts — replace polling with EXTI - Basri Erdogan

Mission: Configure B1 (PC13) as a falling-edge interrupt so that one valid press toggles LD2 (PA5) — without reading the button anywhere in `while (1)`.

Prepare before class

- ☐ Re-read the theory slides up to "The complete interrupt path."
- ☐ Bring your working L05 HAL GPIO project.
- ☐ Know the answer to: why does pressing B1 produce a *falling* edge?

Learning objectives

By checkoff time, you can:

1. configure an EXTI pin and enable its NVIC request in CubeMX;
2. prove with a breakpoint that the interrupt path reaches your callback;
3. pass an event from the callback to `main()` with a `volatile` flag;
4. debounce the button so one press produces exactly one event.

Hardware and files

Item	Required value
Board	Nucleo-F401RE (no wiring — B1 and LD2 are on the board)
Input	B1 on PC13, external pull-up, **low while pressed**
Output	LD2 on PA5, high turns the LED on
Starting point	Copy of your L05 project, saved as `L06_Button_Interrupt`

Configure the interrupt in CubeMX

Open the `.ioc` file. Keep PA5 as `GPIO_Output` (label `LD2`). Then:

1. Click `PC13` and select `GPIO_EXTI13`.
2. In **System Core → GPIO**: falling-edge interrupt mode, **no pull-up and no pull-down** (the board's external resistor already holds PC13 high).
3. In **System Core → NVIC**: enable **EXTI line[15:10] interrupts** (PC13 is line 13; lines 10–15 share this entry). Leave the priority at its default.
4. Generate code, then verify the generated setup in `MX_GPIO_Init()`:

```
GPIO_InitStruct.Mode = GPIO_MODE_IT_FALLING;
GPIO_InitStruct.Pull = GPIO_NOPULL;
HAL_NVIC_EnableIRQ(EXTI15_10_IRQn);
```

Write code only inside `USER CODE` blocks.

Checkpoint A - The interrupt fires

Add below `main()` in the user-code section for functions:

```
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == GPIO_PIN_13) {
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
}
```

Build, flash, press B1 — the LED responds without any button code in the loop.

Evidence for Checkpoint A: a breakpoint on the `if` line is hit when you press B1, and `GPIO_Pin` equals `GPIO_PIN_13`. If execution never stops, recheck the PC13 mode and the NVIC checkbox.

Checkpoint B - The callback reports, the loop acts

The LED action does not belong in the callback. In `USER CODE BEGIN PV`:

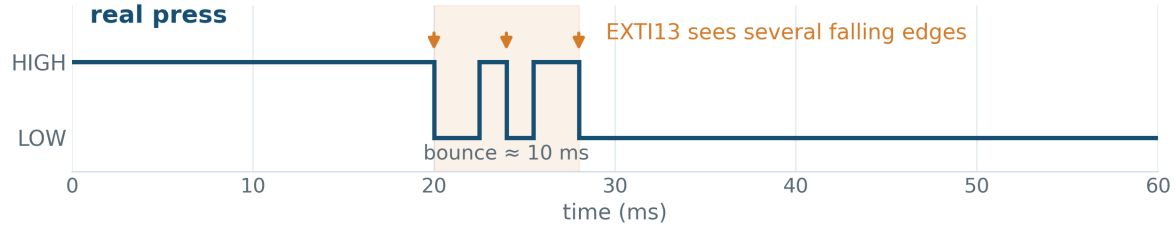
```
volatile uint8_t button_event = 0;
```

Change the callback body to only record the event (`button_event = 1;`), and let the main loop consume it:

```c
if (button_event == 1) {
    button_event = 0;               /* consume before acting */
    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
}
```

Checkpoint C - One press, one event

A real contact bounces — several falling edges per press:



Add `uint32_t last_press_ms = 0;` beside the flag and accept a press only when at least 80 ms has passed since the last accepted one:

```c
uint32_t now = HAL_GetTick();
if ((now - last_press_ms) >= 80) {
    last_press_ms = now;
    button_event = 1;
}
```

The subtraction stays correct even when the millisecond counter wraps.

Test behavior, not just compilation

Test	Expected observation
press once slowly	LD2 toggles once
press ten times	LD2 toggles ten times
hold B1 down	no repeated toggling
release B1	no toggle

Record unexpected behavior *before* changing the code.

Individual extension

Toggle LD2 only after **two** valid presses (a counter in the main loop); the interrupt and debounce code stay unchanged.

Acceptance checklist

- ☐ `.ioc` shows PC13 = `GPIO_EXTI13`, falling edge, no pull, NVIC enabled.
- ☐ Breakpoint proves `HAL_GPIO_EXTI_Callback` runs on a press.
- ☐ `button_event` is declared `volatile`; the LED action lives in `main()`.
- ☐ Ten deliberate presses produce ten reliable state changes.
- ☐ Individual extension demonstrated.

Individual oral check

Be ready for one question and one live change:

- Why does B1 use a falling edge and no internal pull?
- What runs first: `EXTI15_10_IRQHandler()` or your callback?
- Why must the callback stay short, and why is `volatile` required?
- What would happen without the 80 ms guard?

Submit

Submit the project source and one debugger screenshot showing the breakpoint inside your callback.

If you are stuck

Fix the **first** failed stage along the path B1/PC13 → EXTI13 → `EXTI15_10_IRQn` → handler → callback → `button_event` → main loop. **No callback**: PC13 mode / NVIC checkbox. **Wrong pin**: inspect `GPIO_Pin`. **Flag never seen**: missing `volatile`. **Several toggles**: 80 ms guard / `last_press_ms`. **LED dead**: PA5 output setup.